

Database Management

System

Part = 1

(Notes)

Name = Suman Kushwaha

Roll = 81011622

Subject = Database management System

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github = Kushwaha-Suman

website = KushwahaSuman.com.hp

Unit-1

Database and Database User:

Data:

→ Data is a collection of raw facts that by themselves may not have much meaning.

Information:

→ Information is processed data that has meaning and is useful for decision making.

Database



Raw data



processing



Information

Database:

→ A Database is an organized collection of related data that is stored electronically so it can be easily accessed, managed and updated.

Example,

Student table

ID	name	Age	Department
1	Suman	21	BSC CSI
2	Aditi	19	medical
3	Khushi	19	medical

DBms (Database management System)
→ A Database management System (DBMS) is a software that allows users to create, store, retrieve, update and manage data in a database.

User → DBMS → Database

Eg → MySQL

→ PostgreSQL

→ Oracle

→ SQL server

→ SQLite

What can a DBMS do?

→ Create databases

→ Store data

→ Retrieve data

→ Update data

→ Delete data

→ Search data

→ Backup data

→ Secure data

→ Handle multiple users simultaneously

Advantages of DBMS

- ① Reduces Data Redundancy
- ② Reduces Data Inconsistency
- ③ Easy Searching

select * from student
where Name = "Ram";

- ④ Better Security

Student → view only

Teacher → update marks

Admin → full access

- ⑤ Backup & Recovery

- ⑥ multi-user Access

- ⑦ Data integrity

↳ Rules ensure only valid data is stored

- ⑧ faster Data Access

Application of DBMS

- ① Banking

- ② Hospital

- ③ Universities

- ④ Telecommunication

- ⑤ E-Government

- ⑥ manufacturing

Characteristic of the database approach:

① Self describing nature of a database system

↳ A database system is self-describing because it stores both the actual data & the description of the data (metadata)

metadata → data about data

② Insulation Between Programs and Data (Data abstraction)

→ In database management system (DBMS) applications programs are independent of how the data is physically stored.

→ This is called Program - Data independence or Data Abstraction.

③ Support of multiple views of data

→ Different users can see different views of the same database according to their needs.

④ Sharing of data & multiuser transaction processing

→ A DBMS allows multiple users to access and update the database simultaneously while maintaining consistency and correctness.

→ A transaction is a logical unit of work performed on the database.

eg
Deposit money
Withdraw money
Book a ticket
Update students marks

→ Each transaction should either complete fully or not happen at all.

Actors on the scene:

→ Actors on the scene are the people who directly interact with or use the database system.

→ Each actor has a specific role in designing, managing or using the database.

→ The main actors are:-

- 1) Database Administrator (DBA)
- 2) Database Designer
- 3) End Users
- 4) System Analysts and Application Programmers (Software Engineers)

1 Database Administrator (DBA)

→ A database Administrator is the person who is responsible for the overall management, security, maintenance and performance of the database.

E.g In a bank, the DBA manages customer databases control access permissions and performs regular backups.

2 Database Designer:

→ A database designer designs the structure of the database according to the organization requirements.

Responsibilities:

- identifies data to be stored
- Designs tables and relationships
- Defines keys and constraints
- create ER diagrams and database schemas

Example,

For a college database, the designer decides how Student, Teacher, Course and Department tables are related.

2/ End Users :

→ End users are the people who use the database to perform their daily tasks. They usually do not manage or design the database.

a) Casual End User

↳ A casual end user accesses the database occasionally and performs different types of queries.

b) Naive End User

→ A naive end user performs repetitive task using predefined forms or applications without knowing SQL.

c) Sophisticated End User

→ A sophisticated end user understands databases well and can write complex queries for analysis.

Example

Data analyst

Researcher

d) Standalone User

→ A standalone user uses a personal database application on a single computer.

e.g

A small business owner using Microsoft Access or SQLite to maintain inventory record.

4 System Analysts and Application Programmer (Software Engineers)

System Analysts:

- A System Analysts Studies the organization's requirement and decides what the database System should accomplish.

Responsibilities:

- Gather User requirements
- Analyze business processes
- Design the Overall system

Example:

A System analyst gathers requirements for developing a hospital management system.

Application Programmer:

- An application programmer develops software that interacts with the database.

Example:

A Software engineer develops an online shopping website that stores customer and product information in a database.

Workers Behind the Scene :

→ workers behind the scene are professionals who develop, maintain and support the DBMS itself. The main workers behind the scene are:

- 1) DBMS system Designers and Implementors
- 2) Tool Developers
- 3) operators and maintenance personnel

1 DBMS System Designer and Implementors:

→ These are the professionals who design, develop, and improve the DBMS software.

Examples

Engineers working at Oracle, Microsoft SQL Server, or MySQL who develop and maintain the DBMS software.

2 Tool Developers:

→ Tool developers create additional software tools that help users and database administrators work more easily with the DBMS.

eg

Example:

A developer creating a graphical tool like MySQL Workbench or pgAdmin for designing database and writing SQL queries.

3) Operators and maintenance personnel:

→ These are the people responsible for running, monitoring and maintaining the hardware and software environment where the database operates.

Example:

→ An IT support engineer monitoring a company's database server and performing regular maintenance.

Traditional File System vs DBMS :

Traditional File System	DBMS
1) Stores data in separate files.	1) Stores data in a centralized database.
2) High data redundancy (duplicate data).	2) minimizes data redundancy.
3) Data inconsistency is very common.	3) maintains data consistency.
4) Limited security	4) Strong security with user authentication and permissions.
5) Difficult to share data among multiple users.	5) Supports multiple users simultaneously.
6) Backup and recovery are mostly manual.	6) Provides built-in backup & recovery features.
7) Suitable for small applications.	7) Suitable for small, medium & large applications.

V.V.I

Unit = 1

- Actors on the scene
- workers Behind the scene
- Traditional file system vs DBMS
- characteristics of DBMS approach
- Advantage & applications of DBMS